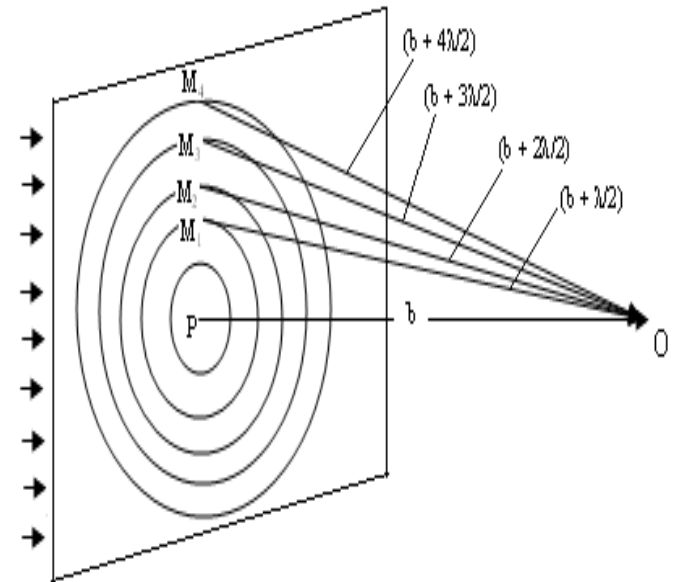


ZONE PLATE

Fresnel zones

- The Fresnel zones are the annular rings in the wave front with the waves from the corresponding points of the adjacent areas differs in path by half wavelength [or reaches the points with a time delay of half time period].
- The areas enclosed between these circles are known as **half period elements or half period zones**.
- The area of the first circle of radius PM is called the first half period zone.
- The areas of the annular rings M_1M_2 , M_2M_3 , etc., are known as second, third, etc., half period zones.



zone plate

- A zone plate is an optical device, which works on the principle of Fresnel's zone.
- A circular aperture in which alternate rings are transparent and opaque is called a Fresnel zone plate
- In Fresnel's zones, the effect at a point due to alternate zones cancels each other.
- If the effect of either even or odd zones is blocked at that point, then the net effect due to alternate zones having a path difference of λ has a maximum value.
- A plate, which allows only one set of alternate zones of the wavefront to pass through it, is called a zone plate.

- The radius of the zones is proportional to *Square root of n* , where $n = 1, 2, 3$, etc.
- This property is used in the construction of a zone plate.
- On a drawing sheet, concentric circles are drawn with the radii proportional to square root of natural numbers as shown in the following figure (1).
- The alternate zones are painted black. A photograph of the pattern is taken. We get on developed negative a reduced pattern.
- This negative forms the zone plate.
- In this plate alternate zones are transparent and allow light and the remaining alternate zones act as opaque region.
- There are two types of zone plates (1) positive zone plate and (2) negative zone plate as shown in figure (1a) & (1b).

- (1) Positive zone plate: A zone plate in which odd zones are transparent and even zones are opaque is known as a positive zone plate.
- (2) Negative zone plate: A zone plate in which even zones are transparent and odd zones are opaque is known as a negative zone plate.



Figure (1a) Positive zone plate



Figure (1b) Negative zone plate

THEORY OF ZONE PLATE

Let AB be an imaginary transparent plate perpendicular to the plane of the paper as shown in figure (2). Let 'S' be a point source of monochromatic light giving out spherical wave of wavelength ' λ '. Draw SP perpendicular to the plane AB and produce it up to 'O'. Let us assume that we want to find out the resultant effect at the point "O" on the screen placed perpendicular to the plane of the paper. With 'P' as centre, imagine the plate AB to be divided into circular half period zones with radii equal to PM_1 , PM_2 , PM_3 , etc., such that;

$$\begin{aligned}
 SM_1 + M_1O &= SP + PO + \lambda/2 \\
 SM_2 + M_2O &= SP + PO + 2[\lambda/2] \\
 &\quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} \\
 SM_n + M_nO &= SP + PO + n [\lambda/2]
 \end{aligned}$$

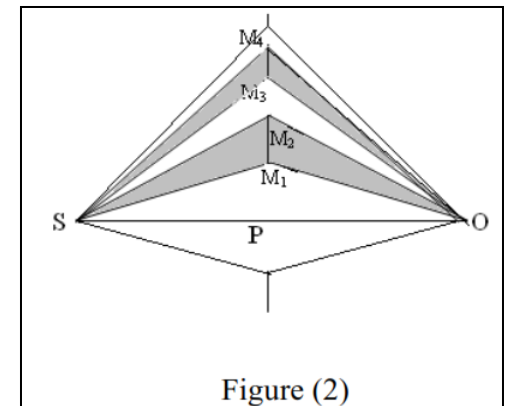


Figure (2)

To determine the radius r_n of the n^{th} circle, consider the relation,

$$SM_n + M_nO = SP + PO + n [\lambda/2]$$

Let $SP = a$, $PO = b$, then we can write,

$$SM_n = \sqrt{SP^2 + PM^2} = \sqrt{a^2 + r_n^2} = a \left[1 + \frac{r_n^2}{a^2} \right]^{1/2} = a + \frac{r_n^2}{2a} \text{ [approximately] } \dots (1)$$

[' r_n ' is very small compared to 'a' and hence by applying binomial theorem and neglecting the square and higher order term, we get the above equation.]

$$\text{Similarly, } M_nO = \sqrt{PO^2 + PM_n^2} = \sqrt{b^2 + r_n^2} = b + \frac{r_n^2}{2b} \dots (2)$$

Substituting for SM_n and M_nO , we get,

$$a + \frac{r_n^2}{2a} + b + \frac{r_n^2}{2b} = a + b + \frac{n\lambda}{2}$$

$$\therefore r_n^2 \left[\frac{1}{a} + \frac{1}{b} \right] = n\lambda$$

$$r_n^2 = n\lambda \left[\frac{ab}{a+b} \right] \text{----(3)}$$

Since a , b and λ are constants, $r_n \propto \sqrt{n}$ ----(4)

It follows that the external radii of the various zones for given values of a and b are proportional to the square root of natural numbers.

$$\text{Area of } n^{\text{th}} \text{ zone} = \pi[r_n^2 - r_{n-1}^2] = \pi \left[\frac{n\lambda ab}{a+b} - \frac{(n-1)\lambda ab}{a+b} \right] = \frac{\pi\lambda ab}{a+b} [n - n + 1] = \frac{\pi\lambda ab}{a+b} \text{ -----(5)}$$

Area of n^{th} zone is independent of the order 'n'. This means that for the given positions of the point source 'S' and its image 'O', the area of all the zones remains practically constant. But, as the amplitude at 'O' m_1, m_2, m_3 , etc., due to various zones decrease very slightly with the order of the zones and are alternately in opposite phase, the resultant amplitude is given by,

$$A = m_1 - m_2 + m_3 - m_4 + \text{----} = \frac{m_1}{2} \text{ [when } n \text{ is very large] -----(6)}$$

If monochromatic light from 'S' is allowed to fall on this zone plate and the emergent light is received on a screen, then for a certain value of 'b', the hypothetical half-period elements of the incident wavefront may coincide with the actual elemental areas of the plate. In such a case if even numbered elements are made opaque, then the resultant amplitude is given by,

$$A = m_1 + m_3 + m_5 + \dots \quad (7)$$

This value is many times greater than that due to all the zones. This means that a zone plate has the focusing action. Under these conditions, 'O' can be said to be the image of 'S'.

Equation (3) can be arranged in the following form.

$$\frac{1}{a} + \frac{1}{b} = \frac{n\lambda}{r_n^2} \quad \dots \quad (8)$$

Comparing equation (8) with the lens formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, -----(9)

$$\text{We have, } \frac{1}{f} = \frac{n\lambda}{r_n^2} \text{ or } f = \frac{r_n^2}{n\lambda} \text{ -----(10)}$$

Equation (10) gives the value of primary or first order focal length of the zone plate.

- Thus , a zone plate acts as a converging lens.
- A zone plate has a number of foci which depend on the number of zones used as well as the wavelength of light employed

Comparison between a zone plate and a convex lens

- Both the zone plate and convex lens form a real image of the object and the equations connecting the conjugate distances are similar
- The focal lengths of both depend on the wavelength and hence suffer from chromatic aberration. The chromatic aberration in a zone plate is much more severe than in a convex lens
- A zone plate acts simultaneously as a convex lens and as a concave lens. In addition to a real image, a virtual image is also formed simultaneously. A convex lens forms only a real image

- In case of zone plate, the image is formed by the diffraction phenomena. In case of a convex lens the image is formed due to refraction of light
- The zone plate has got multiple foci on either side of the plate. Hence, the intensity of the image formed will be much less. Convex lens has only one focus, as all the light is focused at one point, the intensity of the image will be more
- In a zone plate, waves reaching the image point through any two alternate zones differ in path by λ and in phase by 2π . In case of a convex lens, all the rays reaching the image point have zero path or zero phase difference
- A zone plate can be used over a wide range of wavelength from microwaves to x-rays. Glass lens cannot be used beyond the visible region

Comparison between a zone plate and a convex lens:

Zone plate

- (1) Zone plate works on the phenomenon of diffraction
- (2) Odd zone transparent plate produces real image on the opposite side of the plate.
- (3) It shows chromatic aberration as the focal length depends on the wavelength λ
- (4) For a given wavelength λ , zone plate has different focal lengths given by equation

$$f = \frac{r_1^2}{(2p+1)\lambda}$$

- (5) Intensity of the image decreases with the order of focal length
- (6) In a zone plate, the red focus is nearer the plate than the violet focus, hence,
 $f_r < f_v$

Convex lens

- (1) Convex lens works on the phenomenon of refraction of light.
- (2) Convex lens produces real image of the object on the other side of the lens
- (3) It shows chromatic aberration as the focal length depends on the wavelength λ
- (4) A convex lens has a single focal length for a given wavelength given by the formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

- (5) Intensity does not depend on the focal length.
- (6) In a lens the violet focus is nearer than the red focus, hence, $f_v < f_r$

Fresnel diffraction: Circular aperture

- The phenomenon of diffraction usually falls in two categories viz. Fraunhofer and Fresnel.
- Fraunhofer diffraction is an approximation where only plane waves are involved.
- In a diffraction three things are involved, a source, a diffracting element and a screen.
- The diffracting element can be an aperture, a straight edge, an obstacle, a lens, a wire or any other thing which comes in between the source and the screen.

- We observe the intensity pattern on the screen due to the diffracting element.
- Keeping the source and the diffracting element fixed we move the screen towards or away from the diffracting element (and in the next step we move the source keeping the screen fixed) and notice the change in the intensity pattern on the screen

- If there is no **qualitative change** in the pattern apart from a slight scaling, the pattern is identified as Fraunhofer.
- If a qualitative change in the pattern is observed, such as the bright parts of the screen become dark and vice versa or the shape of the pattern changes then we call it a Fresnel diffraction.
- So in the Fraunhofer limit the screen and source both are effectively at a very large (infinite) distance from the diffracting element whereas in the Fresnel case either or both (the source and the screen) are at finite distances from the diffracting element

- Due to diffraction the image of a point source becomes fuzzy on the screen.
- Two nearby points on the object plane (source) may get merged in the image plane (screen) due to the finite size of the diffracting elements (such as lenses etc.) and hence we talk of spatial resolution.
- We are familiar with resolutions of microscopes and telescopes.
- When we view objects through microscopes and telescopes and see their images each point of the object acts as a point source and we see its image little fuzzy.
- Due to diffraction we may not be able to distinguish or resolve the two very close points on the object when we view it through an instrument like microscope or telescope.

Fresnel half period zones

- In the following we first learn a qualitative treatment of Fresnel diffraction due to a circular aperture.
- Later we go for a **quantitative treatment**

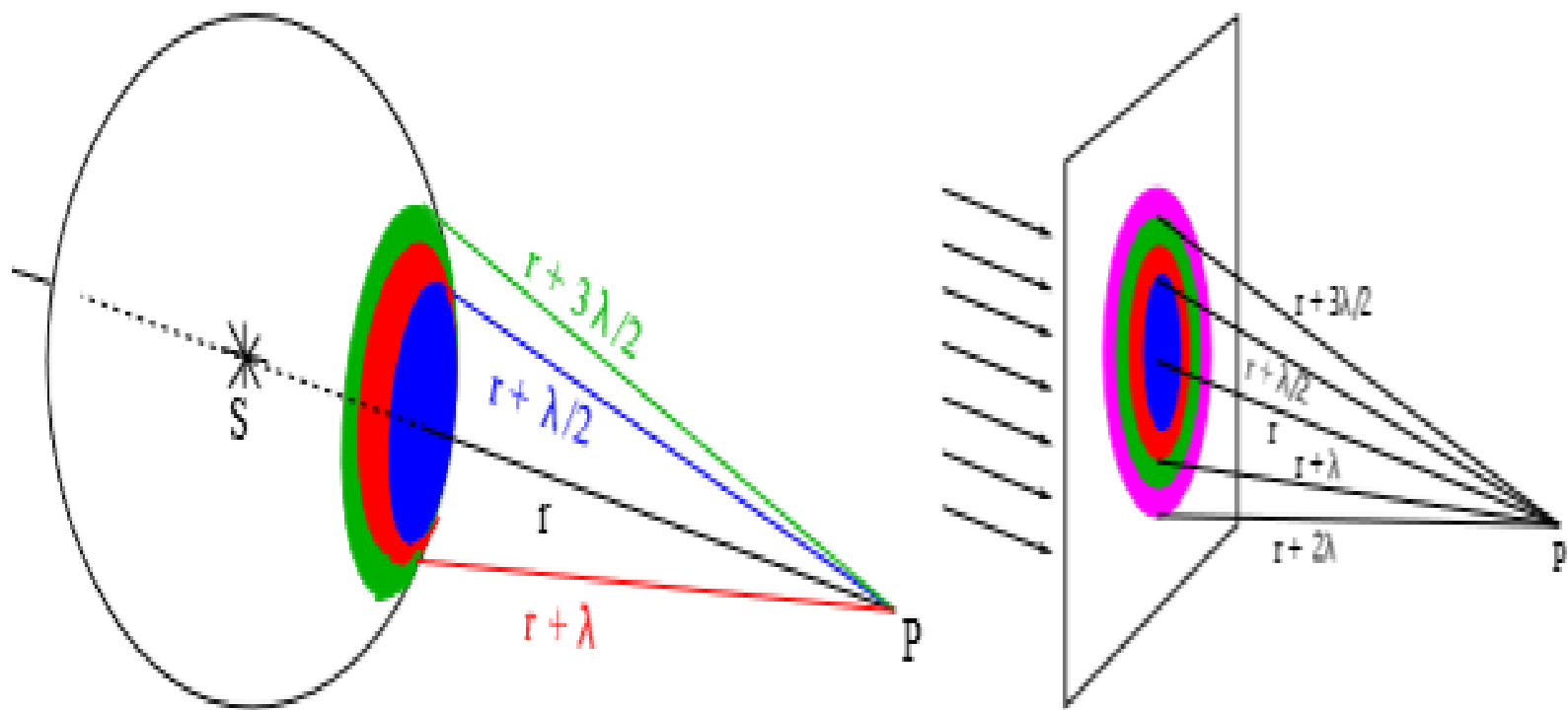


Fig. 24.1: Fresnel zone construction for a point source and a plane wave (source at infinity)

- We refer to the fig 24.1, where Fresnel half period zones (shown with different colours) are created on an arbitrary plane where we intend to put the diffracting element.
- The 'half period' refers to the fact that the path difference between two rays coming from the two consecutive zones and reaching the point of observation, P, is $\lambda/2$ (i.e. half the wavelength of the monochromatic source used).

- The zones are created concentrically around the central one, which is created by the straight line joining the source and the point P.
- Now the rays from one zone and reaching the observation point are (almost) travelling the same distance and hence reach the point P with the same phase.
- Small amplitudes of individual rays from the same zone will then add up to produce a large amplitude at P.
- Whereas two rays coming from two consecutive zones have a path difference of $\lambda/2$ (which corresponds to a phase difference of π) and hence cancel each other.

- We will see in the following that each half period zone has almost the same area so the amplitude of light coming from the different zones are almost the same only their phases are changing alternatively.
- So the contribution from two consecutive zones would produce a null amplitude.
- Now we put a circular aperture on the plane where we constructed the zones.
- The aperture would allow up to certain number of zones and the rest would be cut off

- We count the number of zones allowed by the circular aperture and if that is even then the intensity at P will be almost zero due to the mutual cancellations of the consecutive zones.
- And if the number of zones is odd we have some intensity at the point P and which is equal to the intensity produced by a single Fresnel zone. Let ρ_m be the radius of the m^{th} Fresnel half period zone, then from fig. 24.1

$$r^2 + \rho_m^2 = \left(r + m\frac{\lambda}{2}\right)^2 \quad (24.1)$$

In usual circumstances $\lambda \ll r$ (r is order of meters and λ is order of microns) and hence the λ^2 term can be neglected from the right hand side of the above equation. So we have,

$$\rho_m = \sqrt{mr\lambda}. \quad (24.2)$$

The area, a_m , of the m^{th} zone is given by

$$a_m = \pi(\rho_m^2 - \rho_{m-1}^2) = \pi r \lambda, \quad (24.3)$$

- which is constant and independent of zone number as mentioned earlier.
- So the amplitudes reaching from the different zones are almost the same.
- If one sees more carefully one finds that it decreases slightly with the increase of zone number due to an 'obliquity factor'.
- The situation is depicted in fig 24.2, where the contribution from the first three Fresnel zones are shown in different colours.
- The second diagram shows the situation if one considers semicircular aperture and the observation point is situated above the centre of the circle and it is covering first three Fresnel zones

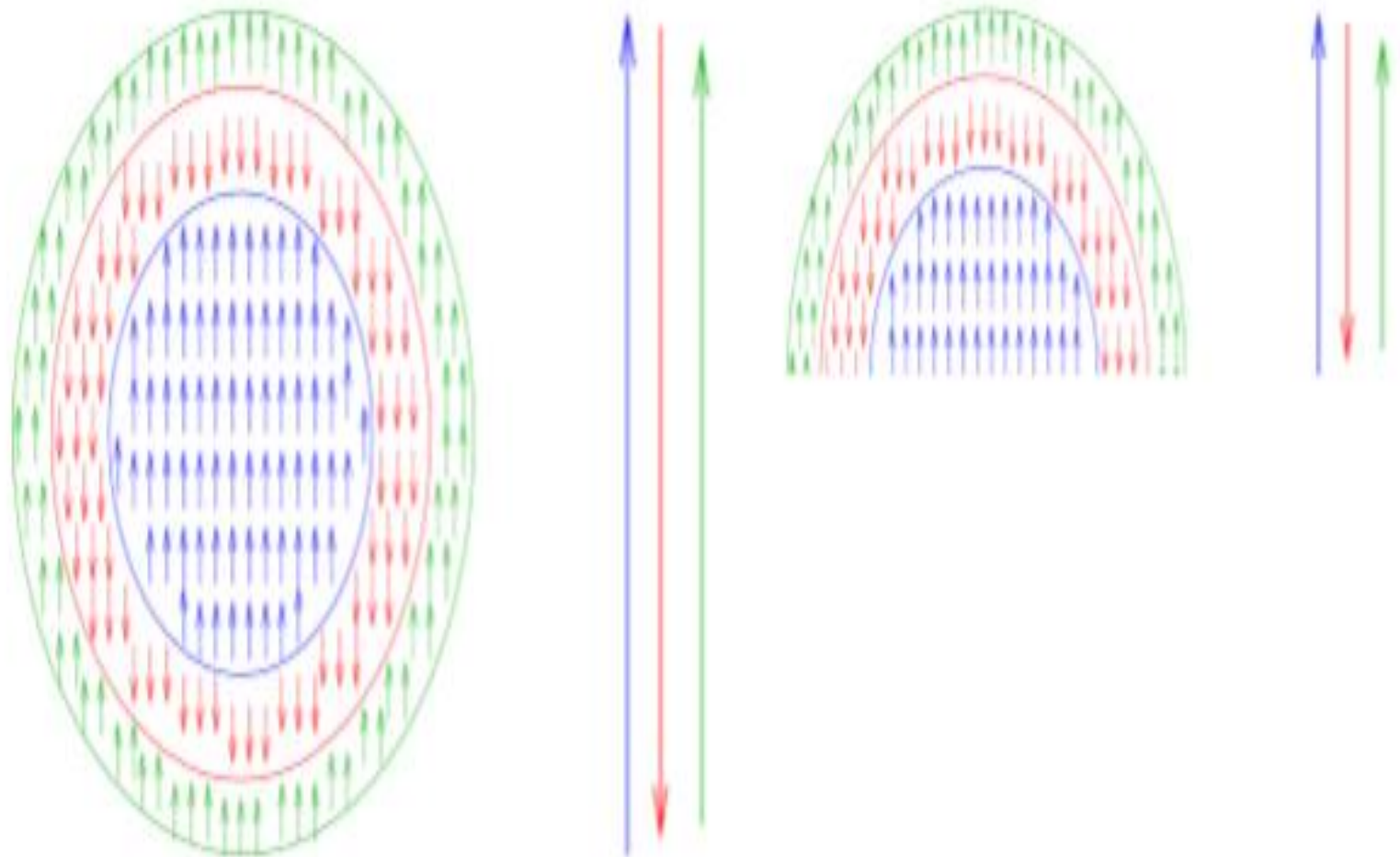


Fig. 24.2: Contribution from the first three Fresnel zones for circular and semi-circular apertures

- In the fig. 24.3 the angle between the ray direction from the source on a certain half period zone (which is also the normal direction of the zone) and the direction of the point of observation from the same zone is θ .
- The obliquity factor, $f(\theta)$, by kirchoff is given by $(1 + \cos\theta)/2$.
- This modifies Huygens secondary wavelets.
- Now with the obliquity factor the Huygens secondary wavelets are not isotropic.
- They will have reduced amplitude as θ increases.
- So, in the backward direction ($\theta = \pi$) the factor becomes zero and hence the amplitude of the secondary wavelets in the backward direction is zero.

- This results in a forward direction propagation of wavefronts and restricts the propagation of wavefronts in the backward direction from a source and which matches with the common observation

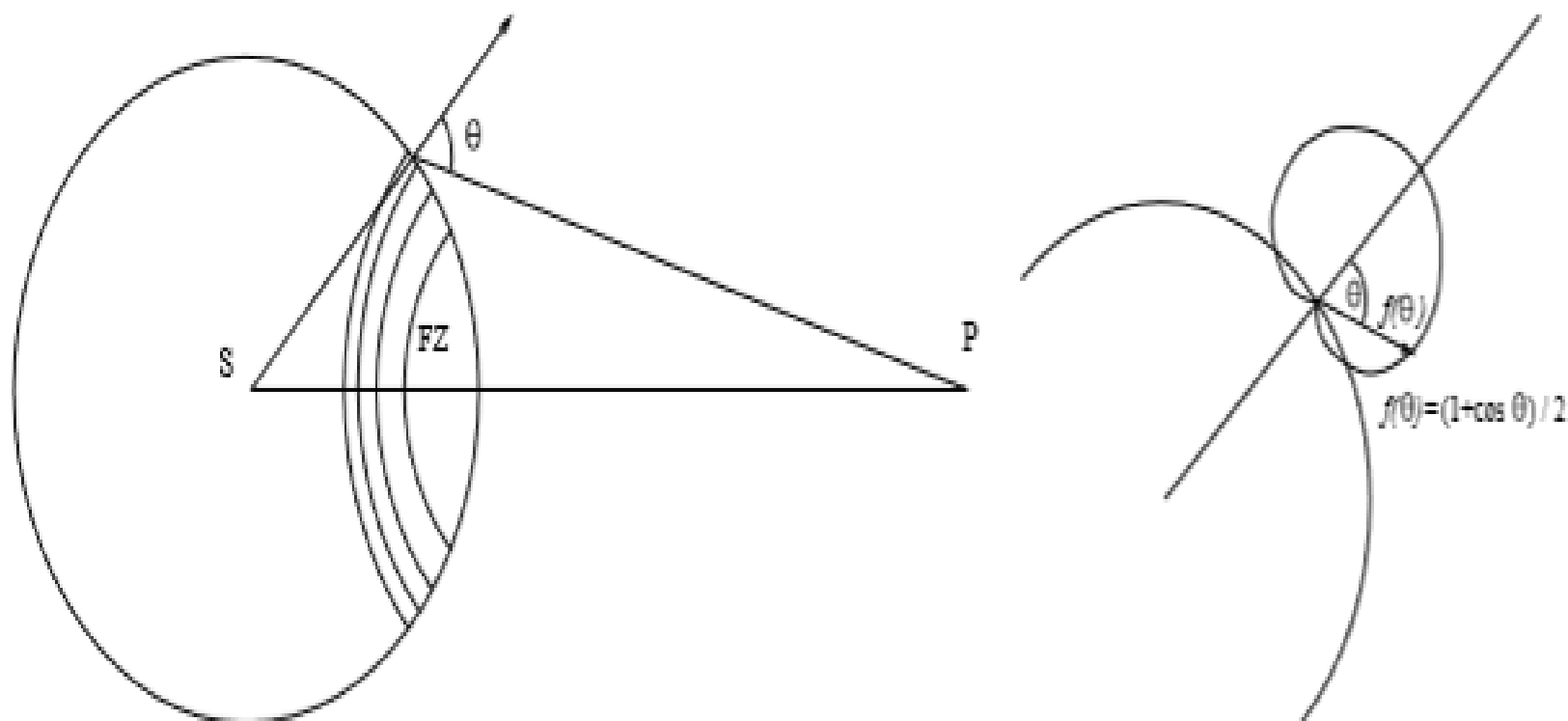


Fig. 24.3: Obliquity factor and Huygens-Kirchoff secondary wavelet

- For simplicity we consider the second diagram of fig 24.1 where the source is at infinity and we have plane waves.
- Now suppose the amplitudes contributing from the consecutive Fresnel zones are $A_1, A_2, A_3 \dots$ etc.
- We would now like to sum these amplitudes for apertures allowing arbitrary number of zones, say first odd m zones.
- The resultant contribution, A , would be,

For simplicity we consider the second diagram of fig 24.1 where the source is at infinity and we have plane waves. Now suppose the amplitudes contributing from the consecutive Fresnel zones are $A_1, A_2, A_3 \cdots$ etc. We would now like to sum these amplitudes for apertures allowing arbitrary number of zones, say first odd m zones. The resultant contribution, A , would be,

$$A = A_1 - A_2 + A_3 - A_4 + A_5 \cdots + A_m. \quad (24.4)$$

The alternate positive and negative signs are due to the π phase difference between the two consecutive zones. The first one is arbitrarily chosen as

positive. The above series can be arranged as the following,

$$A = \frac{A_1}{2} + \left(\frac{A_1}{2} - A_2 + \frac{A_3}{2}\right) + \left(\frac{A_3}{2} - A_4 + \frac{A_5}{2}\right) + \cdots + \left(\frac{A_{m-2}}{2} - A_{m-1} + \frac{A_m}{2}\right) + \frac{A_m}{2}. \quad (24.5)$$

If the terms in the brackets of the equation (24.5) are positive (the amplitude of any zone is less than the average of its adjacent zones, i.e. $(A_{i-1} + A_{i+1})/2 > A_i$) then $A > (A_1 + A_m)/2$. Again we can arrange the same sum in the following way,

$$A = A_1 - \frac{A_2}{2} - \left(\frac{A_2}{2} - A_3 + \frac{A_4}{2}\right) - \left(\frac{A_4}{2} - A_5 + \frac{A_6}{2}\right) \quad (24.6) \\ - \cdots - \left(\frac{A_{m-3}}{2} - A_{m-2} + \frac{A_{m-1}}{2}\right) - \frac{A_{m-1}}{2} + A_m.$$

The above equation says, $A < A_1 - (A_2 + A_{m-1})/2 + A_m$. Since, $A_1 \approx A_2$ and $A_{m-1} \approx A_m$, the above argument shows, $(A_1 + A_m)/2 < A < (A_1 + A_m)/2$, which means

$$A = (A_1 + A_m)/2 \quad (24.7)$$

for odd m . The same argument for even m would conclude

$$A = (A_1 - A_m)/2. \quad (24.8)$$

If the terms of the brackets of the equation (rearrange) are negative (the amplitude of any zone is more than the average of its adjacent zones, i.e. $(A_{i-1} + A_{i+1})/2 > A_i$) then by the same type of arguments as the above, we have $(A_1 + A_m)/2 > A > A_1 - (A_2 + A_{m-1})/2 + A_m$ and the results of

the previous paragraph stay unaltered. When one sums very large number of zones ($m \rightarrow \infty$) as if there were no aperture, then the last zone amplitude, A_m is negligibly small due to the obliquity factor and the total amplitude at the observation point is nothing but the original incident plane wave amplitude A . So we obtain from equation (24.7). So we find, $A_1 = 2A$, that is the amplitude from the first Fresnel zone is twice the incident amplitude and has the same phase as the incident one. Now if there is a circular aperture capturing only the first Fresnel zone for the observation point P, the intensity at P will be four times the incident amplitude.

Thank you